ABDALLAH FEKRY

AI / ML Engineer – Data Scientist – Data Analytics Engineer

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Summary

AI Engineer with hands-on experience in Data Analysis, ML, Deep Learning, NLP, Computer Vision, Agentic AI and RAG Systems, and Deployment techniques. My strong educational background in computer science, combined with multiple courses & internships, has allowed me to build accurate models and successfully lead a variety of projects.

Education

Culture and Science City C.S.C **2020 - 2024**
Bachelor's degree in Computer Science Certificate
Grade: Very Good with Honors
Key Achievements: Graduation Project Publication awarded winner of "Best Distinguished Applied Solution Showcase" at "Computer Vision Projects Expo 2024" Competition by Ready Tensor. Certificate

Skills

Programming: Python, OOP, Data Structures, Structured Query Language SQL
Libraries: NumPy, Pandas, Matplotlib, Seaborn, Plotly, Streamlit UI, Pydantic
ML/DL: Scikit-Learn, TensorFlow, Keras
NLP: LSTM , Lang-Chain, Lang-Graph, Retrieval-Augmented-Generation (RAG), Chroma Database, Gemini API
Dev Tools: Git, Flask APIs, MS Power BI
Methodologies: Agile, Waterfall

Experience

AI Intern | Elite.Org | Remote **October 2026 - Present**

AI Engineer | RaymedAI | Remote **June 2026 – July 2026**
• Architected more than 5 Deep Learning models for healthcare, medical imaging scanning, diseases diagnosis using TensorFlow and Keras. Built Streamlit UI for alpha and Beta test. Deployed models using Flask App and manage API calling with the back-end team to connect with the web app. Designed full documentations for each AI model.

Data Analytics & Agentic AI Intern | Kayfa Academy | Onsite **May 2026 – June 2026**
• Analysed and cleaned large amount of data from several sources as csv, databases, Json, excel sheets, merged them together to build analysis ready model and stored results into MongoDB, designed completed interactive dashboards using Plotly and Streamlit to show insights.
• Developed AI Agents using LangChain, LangGraph, RAG, and Mongo DB to build sales agent that answer business questions, reserved tickets, and recommended courses to the students users. Projects

AI Intern | Electro Pi | Onsite **August 2024 - October 2024**
• Developed more than 7 production ready projects in Computer Vision & LLM using Lang-Chain and RAG automated data extraction from complex JSON structures and created customized agents to answer business questions.
• Applied Computer Vision Techniques as OCR in automated invoicing system embedded with LLM to extract the data from processed 1000+ invoices and add it to a database for the system then deployed it as API using Flask Apps.
• Implemented advanced AI techniques such as Speech Recognition models, Wav2Lip for Lip Syncing combined with LLM agent to build human like chatbots.
• Used Object Detection, extracted image frames from videos, collected datasets, annotated images using RoboFlow, trained Yolo v11 model on the dataset, and enhanced accuracy to exceed +95%.
• Worked individually and as a part of a team using Agile methodology. Certificate

Machine Learning Intern | Code Casa | Remote **September 2023 – October 2023**
• Got hands-on experience on 2 completed projects as Stock Price Prediction using LSTM model architecture and Housing Price Prediction using Multiple Linear Regression. Enhanced their accuracy to +94%.
• Collected, analyzed, preprocessed and cleaned data from 2 datasets then visualized the results for 2 projects.
• Received a Letter of Recommendation after internship for the professional work. LOR - Certificate

AI & IoT Trainee | National Telecommunication Institute NTI **September 2023 – October 2023**
• Trained on AI techniques including mathematics, data preprocessing, cleansing, analysis, and visualization while executed Machine Learning and Deep Learning algorithms and frameworks, and transfer learning. This knowledge enabled the successful completion more than 10 projects.

• Experienced in NLP basic techniques as employed text preprocessing, embedding and models as LSTM in projects as sentiment analysis, and on IMDB classification with accuracy exceeded than 95%.
• Experienced in Computer Vision basic techniques as CNN, RNN, LSTM. Performed 5 projects as chest Xray diagnosis, hand written numbers classification, object detection, face recognition. Improved their accuracy +95% Certificate

Projects

● Smart AI Powered Products Assistant Agent using Lang-Graph and Web Scraping
• Gathered electronical devices data for phones, laptops, and cars from several resources to managed it with Pandas.
• Implemented an agent capable of search in data according to user needs, decided top 3 products and compared between them, recommended top product, then scraped the web stores to get real time prices, links and rate.
WebApp Link GitHub Repo Demo Video

● Smart Doctor and Comprehensive Medical System using Deep Learning
• Led the team and distributed tasks on the team members using Agile methodology, presented work done every meeting weekly.
• Gathered data from several medical datasets as text diseases symptoms for the chatbot module, and images for the medical imaging scan module.
• Architected 17 CNN deep learning models and transfer learned to achieved on average +94% accuracy.
• Deployed models with Flask API to connected with the mobile app, designed interactive UI with Streamlit.
• Publication rewarded winner of "Best Distinguished Applied Solution Showcase" in "Computer Vision Expo" competition by Ready Tensor. Publication Link GitHub Repo Certificate Demo Video

● Interactive Digital Museum using Lang-Chain
• Performed interactive digital museum AI-powered web application that allowed users to communicate with historical characters inside a virtual museum environment.
• Combined conversational AI, voice interaction, visual animations, and computer vision to create an immersive educational experience.
• Designed animations and prompted specific persona for each historical character.
• Deployed projects using Docker file on Google Cloud Run. WebApp Link GitHub Repo Demo Video

● Smart ATS using Lang-Chain and RAG
• Accumulated data from several resources resume pdf and text documents using file scrapper, resume images using OCR model.
• Built a retrieval system that retrieved most similar documents to the job description.
• Embedded an LLM model that collected and reformat the top 5 retrieved documents and the job description then recommend the best candidate that fit the job. GitHub Repo

● Nasa Hazardous Objects Detection using Machine Learning
• Preprocessed the dataset, engineered the features, resolved outliers, null values, imbalanced dataset issues.
• Applied down-sampling and up-sampling techniques together with K-Fold cross validation and ensemble learning models with a non-linear model architecture that enhanced the recall to achieved +98% for the imbalanced dataset.
GitHub Repo

Certificates & Courses

Microsoft Learn Student Clubs by EELU		
• Data Science & Machine Learning Bootcamp	(March 2024 - October 2024)	Certificate
Ministry of Youth and Sports		
• Huawei HCIA AI v3.5 & v4.0	(December 2023) & (Apr 2026)	Certificate
• Data Analysis & SQL Training Program	(July 2023)	Certificate
• Python Programming Language	(October 2022)	Certificate
Hacker Rank		
• Problem Solving Basic	(April 2024)	Certificate
• Python Basic	(October 2022)	Certificate
• SQL Basic	(October 2022)	Certificate
ITI – Mahara-Tech		
• Python Programming Basics	(July 2023)	Certificate
• AI for Everyone (AI4E)	(July 2023)	Certificate
AI Camp & Electro Pi		
• Lang-Chain with RAG course path	(August 2024)	Certificate